

D796

Unix and Linux

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

You are in `/home/ana/projects/web` on a server with 32 GB of RAM. You run `cd ../../docs`. Where are you now?

- A. `/home/docs`
- B. `/docs`
- C. `/home/ana/projects/web/docs`
- D. `/home/ana/docs`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A script has permissions `-rwxr-x---` and is owned by user `deploy` and group `ops`. A contractor in group `ops`, hired last week, tries to use it. What can the contractor do?

- A. It depends on how long the contractor has worked there.
- B. Read, change, and run it.
- C. Nothing at all.
- D. Read and run it, but not change it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which `chmod` value gives the owner read and write, the group read only, and others no access?

- A. 046
- B. 640
- C. 750
- D. 644

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

You want to count the lines in access.log that contain ERROR. The log is 2 GB and rotates at midnight. Which command works?

- A. `grep ERROR access.log > wc`
- B. `wc -l access.log | grep ERROR`
- C. `grep ERROR access.log`
- D. `grep ERROR access.log | wc -l`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A nightly job runs `date > backup.log`. After a week, backup.log has only one line. Why, and what fixes it?

- A. `>` overwrites the file each night; use `>>` to append.
- B. Use `<` instead of `>`.
- C. Use `|` instead of `>`.
- D. Log files can only ever hold one line.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A scheduled job runs `./sync.sh > sync.log`, but error messages still arrive in the admin's email instead of the log. The script is 300 lines long. Which command also captures errors in the log?

- A. `./sync.sh > sync.log 2>&1`
- B. `./sync.sh 2>&1 > sync.log`
- C. Split the script into shorter files.
- D. `./sync.sh >> sync.log`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An admin types `APP_ENV=production` at the prompt, then runs `./start.sh`, which reads `APP_ENV`. The script sees it as empty. The server runs a current Ubuntu release. Why?

- A. That Ubuntu release does not support environment variables.
- B. It was not exported, so child processes do not inherit it; use `export APP_ENV=production`.
- C. Variables set at the prompt never reach any program.
- D. Variable names must be lowercase.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

After running `export APP_ENV=production`, an admin opens a new terminal and finds `APP_ENV` empty. How can the variable be set for every new shell?

- A.** Nothing can make variables last beyond one terminal.
- B.** Run `unset APP_ENV` first.
- C.** Add the export line to a shell startup file such as `~/.bashrc`.
- D.** Run the export command again with `sudo`.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A 40-line script, `deploy.sh`, begins with `#!/bin/bash` and was copied from a teammate's laptop. Running `./deploy.sh` returns "Permission denied." What is the fix?

- A.** `chmod +x deploy.sh`
- B.** Delete the `#!/bin/bash` line.
- C.** Run it with `sudo`.
- D.** Copy it from the laptop again.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

In a shell script, `$?` is 0 right after a command runs. What does that mean?

- A.** The command failed.
- B.** The command succeeded.
- C.** The command produced no output.
- D.** The command is still running.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An admin on a server with 14 user accounts needs a list of every file ending in `.conf` anywhere under `/etc`. Which command fits?

- A.** `grep conf /etc`
- B.** A different command for each user account.
- C.** `find /etc -name '*.conf'`
- D.** `ls /etc`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An admin needs to stop a web server process (PID 4321) and wants it to close connections and save state first. The process was started two days ago. What should the admin do?

- A.** Reboot the whole server.
- B.** Wait until it has run for a week.
- C.** Send kill 4321 (SIGTERM), and use kill -9 only if it does not stop.
- D.** Send kill -9 4321 first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

You are in `/home/ana/projects/web` ~~on a server with 32 GB of RAM~~. You run `cd ../../docs`. Where are you now?

Target word: resulting directory **Decides:** Two levels up from `web`, then into `docs`.

A	Wrong stem number	Goes up three levels instead of two.
B	Confusion set	Treats the relative path as absolute.
C	Dropped element	Ignores the two <code>..</code> segments.
D	Correct	<code>web</code> → <code>projects</code> → <code>ana</code> , then <code>docs</code> .

2. Answer D Concept 2

A script has permissions `-rwxr-x---` and is owned by user `deploy` and group `ops`. A contractor in group `ops`, ~~hired last week~~, tries to use it. What can the contractor do?

Target word: group permissions **Decides:** The group set is `r-x`.

A	Camouflage	Permissions do not check hire date.
B	Confusion set	Those are the owner's rights.
C	Adjacent category	Those are the rights of others, not the group.
D	Correct	Group has read and execute, no write.

3. Answer B Concept 2 · transfer

Which `chmod` value gives the owner read and write, the group read only, and others no access?

Target word: build the octal value **Decides:** `rw-` = 6, `r--` = 4, `---` = 0.

A	Inverted operation	Reverses the owner and others digits.
B	Correct	6 (<code>rw</code>), 4 (<code>r</code>), 0 (<code>none</code>).
C	Confusion set	Adds execute for owner and group.
D	Dropped element	Drops the “others: no access” requirement.

4. Answer D Concept 3

You want to count the lines in `access.log` that contain `ERROR`. ~~The log is 2 GB and rotates at midnight~~. Which command works?

Target word: combine two commands **Decides:** `grep`'s output must feed `wc`.

A	Confusion set	Writes the matches into a file named <code>wc</code> ; nothing is counted.
B	Inverted operation	Counts all lines first, then searches the count.
C	Dropped element	Lists matches without counting them.
D	Correct	The pipe sends matching lines to <code>wc</code> for counting.

5. Answer A Concept 3 · transfer

A nightly job runs `date > backup.log`. After a week, `backup.log` has only one line. Why, and what fixes it?

Target word: overwrite vs. append **Decides:** `>` replaces the file each time.

A	Correct	<code>>></code> adds to the end of the file.
B	Inverted operation	<code><</code> reads input from a file; it does not write.
C	Confusion set	A pipe sends output to a command, not a file.
D	Absolute language	Files can grow; the operator is the issue.

6. Answer A Concept 4

A scheduled job runs `./sync.sh > sync.log`, but error messages still arrive in the admin's email instead of the log. ~~The script is 300 lines long.~~ Which command also captures errors in the log?

Target word: redirect stderr to the file **Decides:** Only stdout is redirected now.

A	Correct	Stdout goes to the file, then stderr follows stdout there.
B	Inverted operation	Wrong order: stderr is pointed at the old stdout before the file redirect.
C	Camouflage	Length does not decide where errors go.
D	Dropped element	Appends stdout but still leaves stderr out.

7. Answer B Concept 5

An admin types `APP_ENV=production` at the prompt, then runs `./start.sh`, which reads `APP_ENV`. The script sees it as empty. ~~The server runs a current Ubuntu release.~~ Why?

Target word: why the child cannot see the variable **Decides:** The variable was not exported.

A	Camouflage	All Linux shells support them.
B	Correct	Only exported variables pass to child processes.
C	Absolute language	Exported ones do.
D	Confusion set	Case is allowed; uppercase is the convention.

8. Answer C Concept 5 · transfer

After running `export APP_ENV=production`, an admin opens a new terminal and finds `APP_ENV` empty. How can the variable be set for every new shell?

Target word: make it persist **Decides:** New shells read startup files.

A	Absolute language	Startup files do exactly that.
B	Inverted operation	Removes the variable.
C	Correct	Startup files run in each new shell.
D	Adjacent category	Changes privileges, not persistence.

9. Answer A Concept 6

A 40-line script, `deploy.sh`, begins with `#!/bin/bash` and was copied from a teammate's laptop. Running `./deploy.sh` returns "Permission denied." What is the fix?

Target word: fix the error **Decides:** The file lacks execute permission.

A	Correct	Adds the execute permission the shell needs.
B	Inverted operation	Removes the line that names the interpreter.
C	Adjacent category	Adds root privileges without fixing the missing execute bit.
D	Camouflage	A fresh copy still lacks execute permission.

10. Answer B Concept 6 • transfer

In a shell script, `$?` is 0 right after a command runs. What does that mean?

Target word: read the exit status **Decides:** 0 means success.

A	Inverted operation	Nonzero means failure.
B	Correct	Exit status 0 signals success.
C	Confusion set	Exit status is separate from output.
D	Adjacent category	<code>\$?</code> is set only after it finishes.

11. Answer C Concept 7

An admin ~~on a server with 14 user accounts~~ needs a list of every file ending in `.conf` anywhere under `/etc`. Which command fits?

Target word: locate files by name **Decides:** The search is on file names, recursively.

A	Confusion set	<code>grep</code> searches file contents, not names.
B	Camouflage	User count does not matter.
C	Correct	<code>find</code> walks the tree matching names.
D	Dropped element	Lists one level with no filter.

12. Answer C Concept 8

An admin needs to stop a web server process (PID 4321) and wants it to close connections and save state first. ~~The process was started two days ago.~~ What should the admin do?

Target word: correct signal order **Decides:** A clean shutdown is wanted.

A	Adjacent category	Disrupts everything to stop one process.
B	Camouflage	Run time does not affect how to stop it.
C	Correct	<code>SIGTERM</code> lets the process clean up; <code>SIGKILL</code> cannot be caught.
D	Inverted operation	<code>SIGKILL</code> stops it with no chance to clean up.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Absolute vs. relative paths	A path starting with / begins at the root; any other path starts from the current directory, where .. means one level up.	Treating a relative path as if it started at root, or miscounting ..	A current directory and a cd with .. segments.
2	Permission bits	Permissions are three sets (owner, group, others) of read 4, write 2, execute 1; each octal digit is the sum for one set.	Reading the wrong set or reversing the digit order.	An rwx string or a three-digit chmod number.
3	Pipes vs. redirection	sends one command's output into another command; > writes output to a file, replacing it; >> appends.	Using > where a pipe or >> is needed.	Combining commands, or a log that keeps losing lines.
4	Standard error	Errors go to stderr (2), separate from stdout (1); 2>&1 placed after the file redirect sends both to the file.	Redirecting only stdout, or putting 2>&1 first.	Errors still appear on screen or in email after redirecting.
5	Environment variables and export	A shell variable reaches child programs only after export, and lasts beyond the session only if set in a startup file.	Assuming a plain assignment is visible to scripts or new terminals.	A script or new terminal that sees a variable as empty.
6	Running shell scripts	A script needs a shebang line and execute permission; exit status 0 means success, nonzero means failure.	Reaching for sudo, or reading 0 as failure.	Permission denied, or checks on \$?.
7	find vs. grep	find searches for files by name and attributes; grep searches inside file contents.	Using grep to locate files by name.	Whether the target is file names or text inside files.
8	Stopping processes	Send SIGTERM first so the process can clean up; use SIGKILL (-9) only if it will not stop.	Starting with kill -9.	A process to stop gracefully.

Course scope (public description): Unix and Linux offers a comprehensive introduction to these operating systems, focusing on essential skills used in system administration and development roles. This course equips students with the ability to employ the most common commands, navigate the Unix/Linux shell, manage files and directories, configure the shell environment, and create shell scripts to automate routine tasks. This hands-on approach prepares students to competently manage and maintain Unix and Linux systems in real-world applications. There are no prerequisites for this course.